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Transforming Multi-Lateral Wells: Strategic Innovations in TAML-5 for Enhanced Lateral Access and Well Production

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Abstract

In multi-lateral wells utilizing the conventional TAML-5 system, accessing a lateral typically necessitates an extensive rig-based workover. This paper focuses on the innovative development of completion equipment tailored for TAML-5 systems that meet high well integrity standards by isolating junctions while enabling independent access to both laterals throughout the well's lifespan. This capability eliminates the need to remove the completion for lateral access, whether for acid stimulation, zonal isolation, or data collection through logging.

A great approach to address the issue of successfully isolating a specific formation is the utilization of multilateral technology. Additionally, due to the requirement for effective junction isolation, TAML level-5 emerged as a viable solution for this issue. Moreover, the integration of well completion design with intelligent completion enhances reservoir management. Nonetheless, sustaining access to both laterals during the well's lifespan is essential for reducing expensive workovers. Consequently, it became imperative to revise and optimize the TAML-5 junction and completion design.

Although standard TAML level 5 has demonstrated efficacy in addressing well integrity concerns, the implementation of dual-string tubing in these completions brings complexity with operational times and pressure control setups, hence complicating deployment. To address this difficulty, an Advanced Vector Block has been designed and integrated into the well completion, requiring lateral access and intelligent completion. This design enables lateral access and improves deployment efficiency by removing the need for a dual-string configuration during deployment phase.

The new system markedly enhances production capacity by integrating inflow control valves for selective lateral production. It offers options for isolating either the mother bore or the lateral through a pressure isolation sleeve or whipstock, both of which can be deployed via wireline or coiled tubing. This innovation facilitates rig-less intervention in any lateral during the life of the well.

The paper presents a demonstration of a successful field technology implementation to the new system, emphasizing that all design enhancement objectives were met. The improved system decreased TAML-5 deployment time by five rig days, benefiting future workover operations.

A major improvement in functionality is represented by the enhanced TAML-5 completion mechanism. By facilitating rig-less access to both laterals and eliminating the necessity for dual-strings, it preserves

the integrity of the junction while reducing well construction and intervention expenses. This development extends the well's lifespan by reducing workover frequency and improves overall operational efficiency.

Introduction

The oil and gas industry is continually looking for ways to improve production efficiency and operational effectiveness. One significant advancement in this pursuit is the implementation of multi-lateral wells, which allow for enhanced reservoir contact and the capability to produce from multiple zones within a single wellbore. These wells provide numerous benefits over traditional single-bore wells, including increased reservoir access, reduced surface footprint, and long-term cost efficiency. However, despite their advantages, the successful implementation of multi-lateral wells often faces challenges, particularly in accessing lateral branches. This is especially true for systems utilizing TAML-5 (Technology Advancement for Multi-Lateral wells) completions, where the complexities of maintaining lateral access can hinder operational effectiveness (Glasser, 2017).

This paper addresses these challenges by discussing strategic innovations in TAML-5 systems that enhance lateral access and well production. It focuses on the development of advanced completion equipment designed to meet stringent well integrity standards while allowing for efficient interventions. The move to TAML-5 represents a critical step in ensuring well integrity, offering improved zonal isolation to prevent unwanted cross-flow between formations. It also simplifies reservoir management, enabling better decision-making throughout the well's life.

Despite the benefits of TAML-5 systems, conventional approaches often require extensive rig-based interventions to access laterals for maintenance, stimulation, or data collection. These interventions can lead to increased operational costs and extended downtimes, negatively impacting production efficiency. Therefore, there is a pressing need for solutions that facilitate rig-less access to laterals while preserving junction integrity.

This paper aims to showcase innovative developments in TAML-5 completion technologies that enable sustained access to laterals throughout the well's productive life. By focusing on advanced designs and completion techniques that reduce the need for rig-based interventions, we highlight the potential for improved production capabilities and overall efficiency in multi-lateral well management. Through a comprehensive analysis of TAML classification systems, the importance of multi-lateral wells, and specific innovations in completion technology, this paper provides insights into the future of reservoir management in an evolving industry landscape.

Background

The transition to TAML-5 significantly enhanced well integrity by improving zonal isolation and preventing unwanted fluid cross-flow. This advancement elevates reservoir management and maintains lateral access throughout the well's life, facilitating better decision-making. However, conventional TAML-5 systems require rig-based interventions for maintenance or data collection, increasing costs and downtime. As a result, there's a growing demand for a solution that allows rig-less access to laterals while maintaining junction integrity.

TAML Classification: Overview and Implementation Guidelines

The Technology Advancement of Multilaterals (TAML) classification system is an industry standard for categorizing multilateral well completions based on their functionality and junction design. The system consists of six levels, ranging from TAML 1 to TAML 6, with each level representing increasing complexity and capabilities (Alraml, 2023)

TAML Classification Levels

The TAML system offers various levels of well integrity and complexity. Starting with TAML Level 1, which involves drilling a lateral from an open hole main bore, primarily used in consolidated formations but lacking flow control. TAML Level 2 (Fig-1a) involves a cased main bore with an open hole lateral, allowing some flow control but requiring full completion removal for lateral access. Level 3 (Fig-1b) features a cased but uncemented lateral, which is less common due to its lack of hydraulic sealing. TAML Level 4 (Fig1-c) includes both cased and cemented main and lateral bores, relying on cement quality for integrity. Level 5 (Fig-1d) provides junction integrity via completion equipment, often used where gas migration is a concern. Finally, TAML Level 6 (Fig-1f) achieves both mechanical and pressure integrity with sealed junctions, offering the highest level of well integrity.



Figure 1a—TAML Level 2.

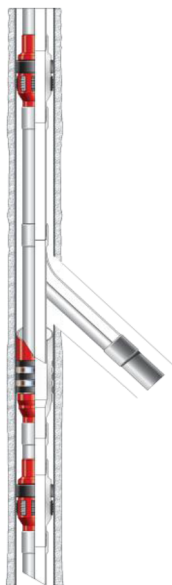


Figure 1b—TAML Level 3.

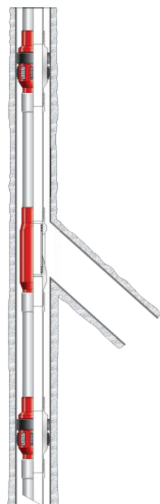


Figure 1c—TAML Level 4.

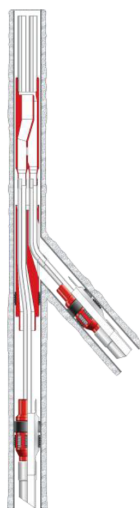


Figure 1d—TAML Level 5.



Figure 1e—TAML Level 6.

Importance of Multi-Lateral Wells

Multilateral wells offer several advantages over traditional single-bore wells. Increased reservoir contact is achieved by drilling multiple laterals from a single wellbore, enhancing production rates by accessing more reservoir area. Reduced Surface Footprint is another benefit, as fewer wells are needed, which is particularly advantageous in environmentally sensitive areas. Despite the potentially complex initial setup, multilateral wells provide Cost-Efficiency through reduced drilling and operational costs over time. However, their success depends on effective access and management of each lateral throughout the well's life.

Selecting TAML-5: Benefits and Considerations. In reservoirs where cemented junctions fall short, TAML-5 offers an optimal solution. It can be integrated into existing TAML-4 or TAML-2 setups if the junction remains open-hole. Unlike traditional cement-based methods, TAML-5 achieves pressure integrity through specialized completion equipment tailored to specific reservoir needs, ensuring complete formation isolation at the junction.

TAML-5 allows access to both the main bore and lateral throughout the well's life, marking a significant advancement in multilateral technology. Recent developments enable independent lateral access, focusing on junction isolation and reducing workovers. This enhances production capabilities, lowers costs, and improves efficiency, making TAML-5 a substantial leap forward in multilateral well management (Benson, 2017).

Legacy Methods

Implementing well integrity measures was vital for development and problem-solving. The approach was tailored to available resources and technological capabilities, ensuring effective solutions within existing infrastructure and cutting-edge advancements. By utilizing available tools and expertise, integrity challenges were efficiently addressed, enhancing well performance and safety.

Hydraulic Isolation and Well Integrity in TAML-5 Completions

The implementation of TAML-5 (Technology Advancement for Multi-Laterals Level 5) completions to achieve hydraulic isolation and maintain well integrity involves a process centered around the deployment of a dual packer system. This approach, while effective, presents both advantages and challenges that merit careful consideration (Fig-2).

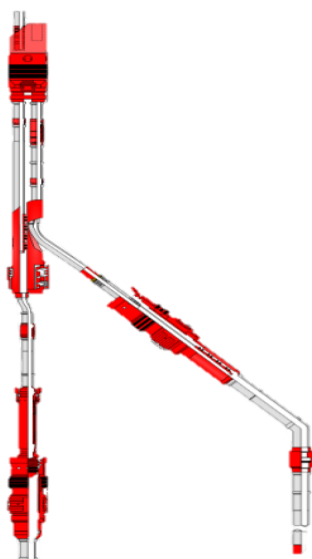


Figure 2—TAML-5 schematic showing the main component of the system

Dual Packer Configuration

The dual packer system is crucial in TAML-5 completions, serving multiple functions. It enables the deployment of two tail pipes: one extending into the lateral to connect with the lateral packer, and the other interfacing with the dual bore deflector in the main wellbore. Above the packer, the design offers flexibility by allowing either dual tubing strings to be run to the surface or integrating them into a single string.

Challenges and Limitations

Despite its effectiveness, the dual packer system poses several challenges. Operational Complexity is increased due to a more time-consuming installation process compared to simpler designs. Running dual packers with dual tubing strings also elevates operational risks. Additionally, Cost Implications are significant, as the system's complexity leads to higher initial completion costs and potential increases in maintenance and workover expenses over the well's life.

Completion Design Trade-offs in legacy design of Multilateral Wells

The choice between dual string and single tubing completions in legacy design of multilateral wells presents distinct advantages and limitations, particularly regarding well access and the implementation of intelligent completion systems.

Dual String Configuration

This system offers several advantages, including access to both the main bore and lateral sections, which enables more flexible well interventions and production management. However, it also has limitations. It restricts the deployment of intelligent completion systems and can result in a more complex and potentially costly completion design (Fig-3b).

Single Tubing Configuration

This system offers several advantages, including the ability to integrate intelligent completion systems and a generally simpler completion design compared to dual string setups. However, it also has limitations. Access is primarily restricted to the main bore, and accessing the lateral requires retrieving the entire completion, which results in increased downtime and operational costs for interventions (Fig-3a).



Figure 3a—TAML-5 with dual packer & dual string to surface



Figure 3b—TAML-5 with dual packer & single string to surface

Industry Improvement

The trade-offs in this completion technique underscore ongoing challenges in multilateral well design. Legacy TAML-5 completions with dual packers offer significant benefits in zonal isolation and integrity, but the industry seeks innovations for more efficient installations and flexible interventions. Future developments may focus on simplifying the installation process, reducing risks, and enhancing independent access to different well sections, potentially addressing the current limitations of the dual packer system.

Technology Overview

Advancements in multilateral well technology hinge on optimizing operational sequences, mitigating risks, and integrating access capabilities with intelligent completion systems. This approach enhances efficiency, safety, and overall well performance. Key equipment and technologies have driven these improvements, contributing to the evolution of multilateral well operations and more effective, economical management strategies.

Dual Bore Deflector

The dual bore deflector is a crucial component in multilateral well completions, primarily serving two purposes: guiding tools into the lateral wellbore and maintaining pressure isolation between the main wellbore and the lateral. It features two separate flow paths, a sealing mechanism for pressure integrity, and orientation capabilities to direct tools accurately into the lateral. Installation typically occurs after setting the lateral packer and retrieving the whipstock, often in conjunction with other completion equipment. The deflector facilitates dual-string configurations, enabling independent control and production from both the main and lateral wellbores (Fig-4).



Figure 4—Dual bore Deflector

Vector Block

The multilateral vector block represents a significant advancement in well completion technology, integrating intelligent completion systems while maintaining well access throughout its lifespan. Its innovative design features a twin-flow mechanism, separating lateral and main bore flows, which is crucial for enhanced reservoir management and production optimization. The vector block is versatile, compatible with various accessories, and adaptable to changing conditions.

It serves three main purposes: integrating intelligent completion technology with multilateral technology in TAML-5, enabling selective access to individual laterals, and facilitating single-string commingled flow from multiple laterals. Key features include multiple flow paths for commingled production, selective access without pulling the upper completion, and compatibility with TAML Level 5 junctions.

The benefits include efficient production through commingled flow, improved intervention capabilities, and enhanced reservoir management when used with intelligent completions, allowing for monitoring and flow control of each lateral branch (Fig-5).



Figure 5—Vector Block with pressure isolation sleeve and tubing exit whipstock

Intervention Methodology

The Vector block represents a significant advancement in multilateral well technology, designed to facilitate independent flow from each lateral while maintaining well access throughout its operational life without

the need for completion retrieval. This innovative design incorporates two key components that enhance intervention capabilities:

Pressure Isolation Sleeve

This advanced tool is engineered to be installed within the Vector block, effectively isolating the lateral while maintaining access to the main bore. Its design features top and bottom seals that engage with designated seal areas within the block. The sleeve is securely anchored via a collet mechanism that locks into a specially designed profile inside the Vector block, ensuring reliable isolation and easy retrieval when needed (Fig-6).



Figure 6—Pressure Isolation Sleeve set inside the Vector Block

Tubing Exit Window

This component is a critical element for lateral access. Designed for precise installation within the Vector block, it features a carefully engineered deflector. When deployed, this deflector redirects the Bottom Hole Assembly (BHA) towards the lateral. The deflector's orientation is meticulously calibrated to guide tools accurately into the lateral, ensuring efficient and reliable access for various intervention operations (Fig-7).



Figure 7—Tubing Exit Window set inside the Vector Block

Big bore packer

The big bore packer design represents a development in multilateral well completion technology. It is constructed to house the Vector block and associated tailpipe, while maintaining a large internal diameter and incorporating seal areas that facilitate integration with intelligent completion components. The packer utilizes a hydraulic setting mechanism and is deployed using a specialized setting tool. This design approach combines elements of conventional completion methods with more advanced intelligent systems. The packer aims to provide effective sealing, maintain flow capacity, and contribute to well control in multilateral applications, potentially enhancing reservoir management capabilities.

Technological Convergence of Multilateral Wells and Intelligent Completion

Enhancing multilateral well completions with advanced production control capabilities has significantly improved reservoir management strategies. Integrating intelligent completion systems, designed to regulate each lateral independently, required extensive equipment customization. This approach involves several key aspects: Flow area optimization, which tailors flow areas to match reservoir characteristics and production goals, ensuring optimal flow rates and pressure management for each lateral. Equipment Compatibility is also crucial, involving the development of components that integrate seamlessly with existing multilateral technologies and ensuring interoperability between ICVs and other downhole equipment. Additionally, Strategic ICV Placement determines optimal locations based on reservoir geology, well trajectory, and production objectives to maximize control and efficiency. Finally, Comprehensive Monitoring Systems are implemented using advanced sensing technologies to monitor all potential production zones, with real-time data acquisition systems for continuous monitoring.

Technology overview

Intelligent completion systems significantly improve reservoir management by providing active monitoring, predictive analytics, and informed decision-making. These systems optimize production, manage water output, minimize environmental impacts, and reduce costs. They are particularly effective in complex wells

such as horizontal wells with geological heterogeneity, multizone producers, multilateral wells, and wells with ESPs.

Key Components

- **Interval Control Valves (ICVs):** Regulate fluid flow or injection, configurable as open/close or choking valves.
- **Permanent Downhole Measuring Gauges (PDHMGs):** Provide real-time pressure and temperature data, essential for monitoring well conditions. Offer single, dual, or triple sensor configurations for comprehensive wellbore monitoring.
- **Feedthrough Isolation Packers:** Enable individual zone control with multiple feedthroughs for control lines.
- **Surface Control Panels:** Manage ICVs and interface with surface hydraulic systems.
- **Shrouded ICV**

A shrouded ICV (Interval Control Valve) is a specialized type of flow control valve used in intelligent well completions, particularly in multilateral. Key features of shrouded ICVs include:

1. **Protective shroud:** The valve is encased in an outer sleeve or shroud, creating a separate flow path for production from different zones.
2. **Flow redirection:** The shroud allows production from lower zones to flow through a dip tube and the shroud before entering the production tubing through the ICV.
3. **Integration with other technologies:** The shrouded design allows for seamless integration with sand control, multilateral systems, and other completion technologies.

Deploying the New TAML-5 System: Key Considerations and Processes

Initial Drilling and Preparation.

- The main bore is drilled to the desired depth.
- The lower completion is installed within the main bore.
- A window is created in the casing to allow for lateral access.
- The lateral section is then drilled to extend into the reservoir.

Lower Completion Installation.

- The lower completion is deployed and securely set in place.
- The lateral packer is installed and properly set (see [Fig-8a](#)) to ensure isolation.

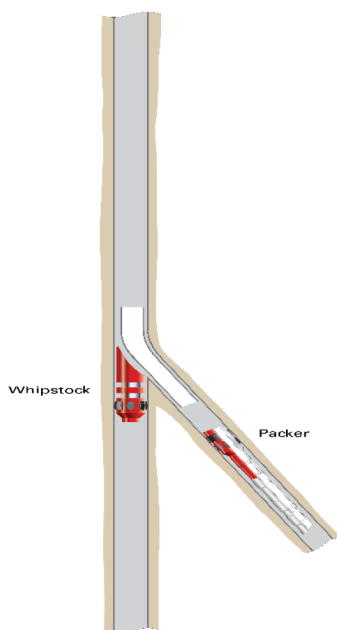


Figure 8a—Lateral Packer installed and tested

Whipstock Retrieval and Packer Deployment.

- The whipstock is retrieved from the wellbore (see [Fig-8b](#)) to prepare for the next steps.
- The lower main bore packer is deployed and subsequently pressure-tested to ensure integrity.

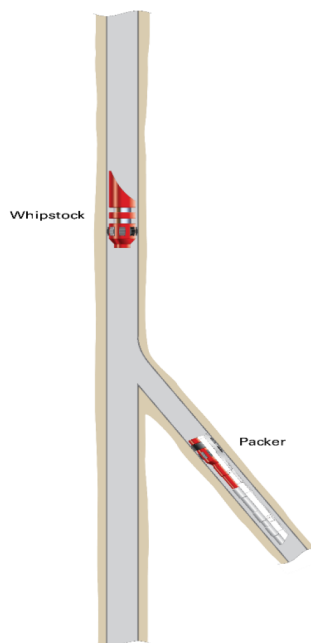


Figure 8b—Lateral Packer installed and tested

Dual Bore Deflector Installation.

- The dual bore deflector is lowered into the well.
- It is positioned correctly, set in place (see [Fig-8c](#)).

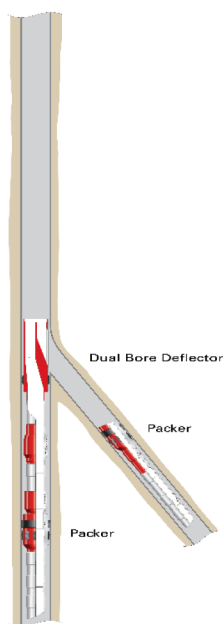


Figure 8c—Dual Bore Deflector installed and tested

Vector Block and Tailpipe Installation.

- The vector block is deployed into the system (see [Fig-8d](#)) to facilitate flow management.
- Two tailpipe strings are then run and secured:
 - One string is set within the lateral packer.
 - The other string is set in the dual bore deflector, ensuring proper connectivity.

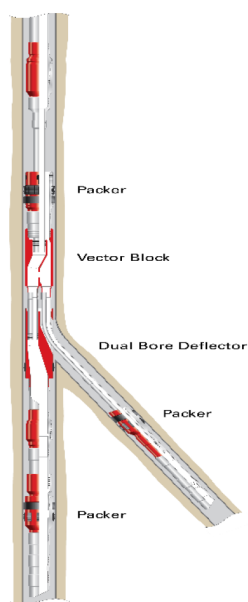


Figure 8d—Vector Block installed and tested

Upper Completion Installation.

- The upper completion string, featuring intelligent completion components, is deployed and set. (Fig-9)

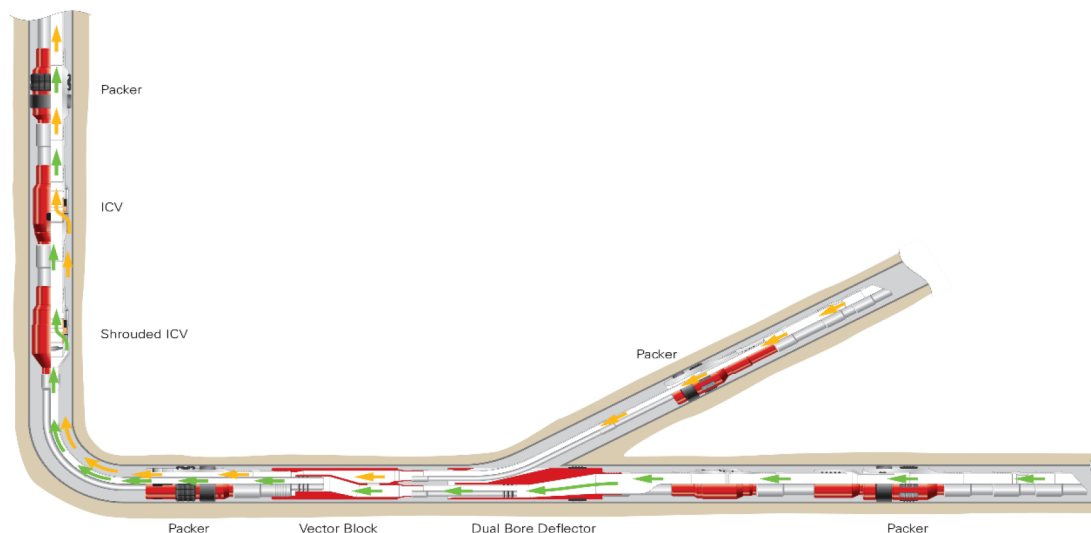


Figure 9—Schematic shows the whole completion and flow direction

Enhanced Value Provided by the New TAML-5 System

Addressing Formation Isolation

To effectively access specific formations within multi-lateral wells, advanced multilateral technology is crucial. The TAML-5 level proves to be a viable solution due to its capability to isolate junctions efficiently. The design process included a thorough analysis of existing completion technologies, identifying gaps, and refining the junction and completion design to ensure consistent access to both laterals.

Integration of Intelligent Completion

Incorporating intelligent completion systems into TAML-5 designs significantly improves reservoir management by providing real-time data and enabling precise control over production parameters. This integration is essential for maintaining optimal reservoir conditions and ensuring long-term productivity.

Benefits of Intelligent Completion

- **Real-Time Monitoring:** Sensors within the completion continuously measure pressure, temperature, and flow rates, allowing to make informed decisions.
- **Automated Control Systems:** These systems adjust production rates from each lateral based on real-time data, optimizing recovery while minimizing reservoir depletion.
- **Data Analytics:** Advanced analytics identify trends and anomalies, facilitating proactive maintenance and intervention strategies.

Design Optimization

Revising and optimizing the TAML-5 junction and completion design were crucial for achieving the desired functionality, which included the development of an Advanced Vector Block that facilitates lateral access without the complexities associated with dual-string tubing configurations, as well as the integration of inflow control valves that allow for selective production from each lateral, thereby enhancing overall production capacity.

Advanced Vector Block Design

The Advanced Vector Block features.

- **Streamlined Installation:** Designed for easy setup, reducing the need for complex rig arrangements.
- **Enhanced Flow Dynamics:** Optimized internal geometries promote efficient fluid movement and minimize pressure losses.

Rig-less Intervention Capabilities. The new completion design includes mechanisms that enable rig-less intervention in any lateral throughout the well's life. This capability is achieved through:

- **Pressure Isolation Sleeves:** These sleeves isolate either the mother bore or the lateral, allowing for independent access without compromising overall well integrity.
- **Whipstock Technology:** This enables re-entry into the lateral without extensive rig setups.

Benefits of Rig-less Interventions.

- **Reduced Downtime:** Rig-less access significantly shortens the time required for maintenance and stimulation operations.
- **Cost Savings:** New system reduce expenses related to rig use, transportation, and logistics associated with traditional workover methods.

Transformative Outcomes.

- **Operational Efficiency:** The implementation of the advanced TAML-5 completion system has significantly improved operational efficiency. Data collected from field trials indicate a reduction in deployment time by five rig days, translating to substantial cost savings and enhanced production timelines.
- **Enhanced Production Capacity:** The integration of inflow control valves has provided the industry with the flexibility to optimize production rates from each lateral independently. This selective production capability has resulted in increased overall well output and better reservoir management.
- **Successful Field Demonstrations:** Field demonstrations of the new system have confirmed that all design enhancement objectives were met. The ability to conduct rig-less interventions has proven invaluable, allowing for timely maintenance and optimization activities without the logistical challenges associated with traditional workover methods.

Conclusion

The advances in TAML-5 systems for multi-lateral wells mark a significant progress in well completion technology. By facilitating independent access to laterals and enhancing production capacity without compromising junction integrity, these innovations present a strong case for their broader adoption in the oil and gas industry.

Key highlights include:

- **Improved Operational Efficiency:** The new TAML-5 completion systems have demonstrated a notable reduction in deployment time, allowing for faster access to laterals and minimizing downtime.
- **Increased Production Capacity:** The integration of inflow control valves enables optimizing production rates from each lateral independently, resulting in higher overall output.

- **Successful Field Demonstrations:** Field trials have validated the effectiveness of the new systems, achieving high success rates for rig-less interventions and significant cost savings compared to traditional methods.
- **Sustainable Practices:** These advancements pave the way for more sustainable well management practices, enhancing productivity while reducing operational costs.

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